**Project Design Phase-II**

**Technology Stack (Architecture & Stack)**

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|---|---|
| Date | 29 September 2026 |
| Team ID | SWTID-2026-1637 |
| Project Name | AI Study Buddy |
| Maximum Marks | 4 Marks |

**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

**Reference:** https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

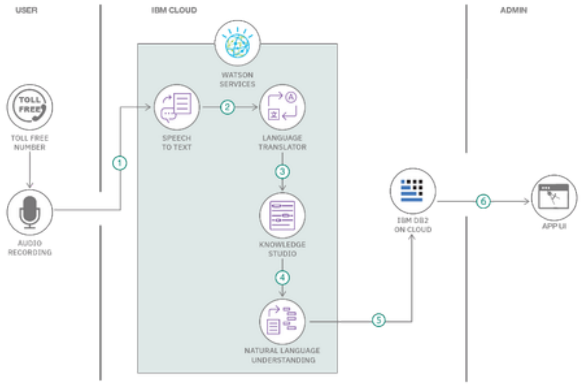



**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | How user interacts with application e.g.<br><br>Web UI, Mobile App, Chatbot etc. | HTML, CSS, JavaScript / Angular Js / React Js etc. |
| 1. | Application Logic-1 | Logic for a process in the application | Java / Python |
| 1. | Application Logic-2 | Logic for a process in the application | IBM Watson STT service |
| 1. | Application Logic-3 | Logic for a process in the application | IBM Watson Assistant |
| 1. | Database | Data Type, Configurations etc. | MySQL, NoSQL, etc. |
| 1. | Cloud Database | Database Service on Cloud | IBM DB2, IBM Cloudant etc. |
| 1. | File Storage | File storage requirements | IBM Block Storage or Other Storage Service or Local Filesystem |
| 1. | External API-1 | Purpose of External API used in the application | IBM Weather API, etc. |
| 1. | External API-2 | Purpose of External API used in the application | Aadhar API, etc. |
| 1. | Machine Learning Model | Purpose of Machine Learning Model | Object Recognition Model, etc. |
| 1. | Infrastructure (Server / Cloud) | Application Deployment on Local System / Cloud<br><br>Local Server Configuration:<br><br>Cloud Server Configuration : | Local, Cloud Foundry, Kubernetes, etc. |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | List the open-source frameworks used | Technology of Opensource framework |
| 1. | Security Implementations | List all the security / access controls implemented, use of firewalls etc. | e.g. SHA-256, Encryptions, IAM Controls, OWASP etc. |
| 1. | Scalable Architecture | Justify the scalability of architecture (3 – tier, Micro-services) | Technology used |
| 1. | Availability | Justify the availability of application (e.g. use of load balancers, distributed servers etc.) | Technology used |
| 1. | Performance | Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc. | Technology used |

**References:**


https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d